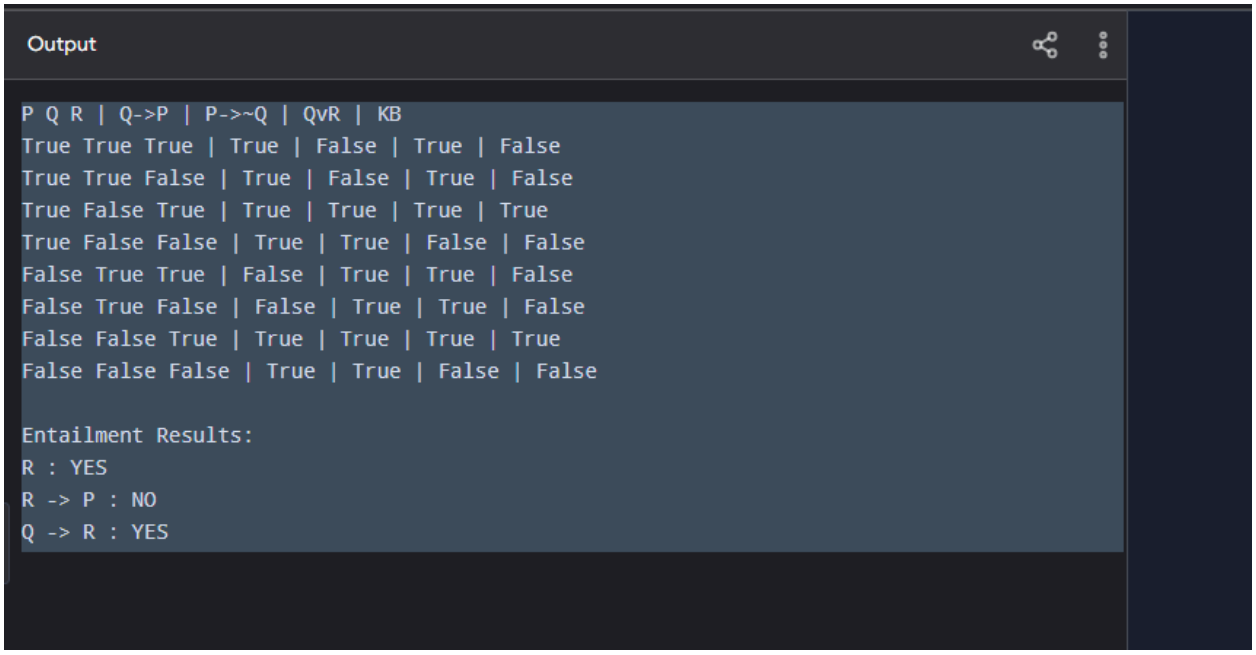
PropositionalLogic_TruthTableEnumeration

```
from itertools import product

def tt_entails():
    symbols = ['P', 'Q', 'R']

    queries = {
        'R': lambda P, Q, R: R,
        'R -> P': lambda P, Q, R: (not R) or P,
        'Q -> R': lambda P, Q, R: (not Q) or R
    }

    results = {q: True for q in queries}

    print("P Q R | Q->P | P->~Q | QvR | KB")

    for P, Q, R in product([True, False], repeat=3):

        s1 = (not Q) or P
        s2 = (not P) or (not Q)
        s3 = Q or R

        KB = s1 and s2 and s3

        print(P, Q, R, "|", s1, "|", s2, "|", s3, "|", KB)

        if KB:
            for name, query in queries.items():
                if not query(P, Q, R):
                    results[name] = False

    print("\nEntailment Results:")

    for query, result in results.items():
        print(query, ":", "YES" if result else "NO")

tt_entails()
```


Output
P Q R | Q->P | P->~Q | QvR | KB
True True True | True | False | True | False
True True False | True | False | True | False
True False True | True | True | True | True
True False False | True | True | False | False
False True True | False | True | True | False
False True False | False | True | True | False
False False True | True | True | True | True
False False False | True | True | False | False
Entailment Results:
R : YES
R -> P : NO
Q -> R : YES